

PHYS 661 Fall 2024

Problem set 5, due November 7 2024. Each problem is 10 pts unless otherwise stated. Submit your homework either (1) via Brightspace or by (2) emailing it directly to the TA at gbowling@purdue.edu.

1 1D δ -potential revisited (15 pts)

Consider a 1D potential well $U(x) = -\alpha\delta(x)$ where $\alpha > 0$ so there is a bound state. The ground state of this system is a bound level with wave function $\psi_0(x) = \sqrt{\kappa}e^{-\kappa|x|}$ and energy $E_0^{(0)} = -\frac{\hbar^2\kappa^2}{2m}$, where $\kappa = m\alpha/\hbar^2$.

1. Show that the continuous spectrum wave functions are $\psi_{+,k}(x) = \frac{1}{\sqrt{\pi}} \cos(k|x| + \frac{\varphi_k}{2})$ and $\psi_{-,k} = \frac{1}{\sqrt{\pi}} \sin kx$, both with energy $E_k^{(0)} = \frac{\hbar^2 k^2}{2m}$. (The subscript $\pm$ denotes spatial parity eigenvalues.) Find the phase φ_k .
2. Show that the continuous spectrum states are δ -normalized, $\int_{-\infty}^{\infty} dx \psi_{a,k}(x) \psi_{b,k'}(x) = \delta_{ab} \delta(k - k')$. Show that they are orthogonal to the bound state ψ_0 . (Note that it is enough to consider $k, k' > 0$ since the states with $k < 0$ are not linearly independent.) You can use the helpful identity $\int_0^{\infty} dx e^{i(k-k')x} = i \frac{1}{k-k'} + \pi \delta(k - k')$.
3. By using a scattering wave function of the form $\psi_k(x) = \begin{cases} e^{ikx} + r e^{-ikx} & x < 0 \\ t e^{ikx} & x > 0 \end{cases}$, find the transmission coefficient $D = |t|^2$ of the scattering state. Express D in terms of the energy of the scattering state.

2 Scattering time-delay (15 pts)

Consider 1D scattering with a Gaussian wave packet so that the incident wave from the left is $\psi_{\text{inc}}(x, t) = \int_0^{\infty} dk e^{-\frac{1}{2}(k-k_0)^2/\Delta k^2} e^{-ikx} e^{-iE_k t/\hbar}$ for $x > a$ and the reflected wave is assumed to be $\psi_{\text{refl}}(x, t) = \int_0^{\infty} dk e^{-\frac{1}{2}(k-k_0)^2/\Delta k^2} e^{2i\delta(k)} e^{ikx} e^{-iE_k t/\hbar}$ for $x > a$. We denote $E_k = \hbar^2 k^2/(2m)$.

1. Show that in the absence of scattering, the reflected wave packet peak at time t is at position $x = v_0 t$ where v_0 is the mean velocity. You can assume Δk is small.
2. Calculate the peak position in the presence of scattering and show that now the peak is at $x = v_0(t - \Delta t)$, where $\Delta t = 2\hbar \left[\frac{d\delta(E)}{dE} \right]_{E=E_{k_0}}$.

3. Calculate the time-delay for the example $V = \alpha\delta(x - a)$ that we discussed in the lecture. In the limit of large barrier, find the time delay for a wave packet that has k_0 such that 1. $|E_{k_0} - E_n| \gg \Gamma_n$, and 2. $E_{k_0} = E_n$. Comment on your findings.

3 Born approximation

By using the Born approximation, find the scattering amplitude and total scattering cross section for the following 3D central potentials. For the total cross section, you can assume low energy and give leading $E \rightarrow 0$ behavior.

1. $V(r) = \alpha\delta(r - a)$
2. $V(r) = \begin{cases} V_0 & r < a \\ 0 & r > a \end{cases}$

4 Scattering off an impenetrable sphere

Consider 3D scattering off a potential $V(r) = \begin{cases} \infty & r < a \\ 0 & r > a \end{cases}$.

1. Find the s -wave ($l = 0$) phase shift.
2. Find the total low-energy ($k \rightarrow 0$) cross section σ . You may use $\frac{d\sigma}{d\Omega} = |f(\theta)|^2$ with $f(\theta) = \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) e^{i\delta_l} P_l(\cos\theta) \sin\delta_l$.